

Strings

LEVEL 1: String Basics

Program 1: Input and Display String Write a program that takes a string as input and displays it.

Program 2: Find Length of String Write a program that calculates and displays the length of a string without using `strlen()`.

Program 3: Count Vowels in String Write a program that counts how many vowels (a, e, i, o, u) are in the string.

Program 4: Count Consonants in String Write a program that counts how many consonants are in the string.

Program 5: Count Spaces in String Write a program that counts how many spaces are present in the string.

Program 6: Count Digits in String Write a program that counts how many digits (0-9) are in the string.

Program 7: Count Alphabets in String Write a program that counts how many alphabetic characters are in the string.

Program 8: Count Special Characters Write a program that counts how many special characters (not alphabet, digit, or space) are in the string.

Program 9: Count Uppercase Letters Write a program that counts how many uppercase letters (A-Z) are in the string.

Program 10: Count Lowercase Letters Write a program that counts how many lowercase letters (a-z) are in the string.

Program 11: Count Words in String Write a program that counts how many words are in the string (words are separated by spaces).

Program 12: Display String Character by Character Write a program that displays each character of the string on a new line.

Program 13: Display String in Reverse Order Write a program that displays the string in reverse order without modifying it.

Program 14: Find First Character Write a program that displays the first character of the string.

Program 15: Find Last Character Write a program that displays the last character of the string.

LEVEL 2: Character Operations

Program 16: Convert to Uppercase Write a program that converts all lowercase letters to uppercase in the string.

Program 17: Convert to Lowercase Write a program that converts all uppercase letters to lowercase in the string.

Program 18: Toggle Case Write a program that toggles case of each character (uppercase ↔ lowercase).

Program 19: Capitalize First Letter of String Write a program that capitalizes only the first letter of the string.

Program 20: Capitalize First Letter of Each Word Write a program that capitalizes the first letter of every word in the string.

Program 21: Replace All Spaces with Underscore Write a program that replaces all spaces with underscores (_).

Program 22: Replace Character X with Y Write a program that replaces all occurrences of character X with character Y.

Program 23: Remove All Spaces Write a program that removes all spaces from the string.

Program 24: Remove Leading Spaces Write a program that removes spaces from the beginning of the string.

Program 25: Remove Trailing Spaces Write a program that removes spaces from the end of the string.

Program 26: Remove Extra Spaces (Keep Single) Write a program that removes extra spaces, keeping only single space between words.

Program 27: Count Occurrence of Character Write a program that counts how many times a specific character appears in the string.

Program 28: Find First Occurrence of Character Write a program that finds the first position where character X appears. Display -1 if not found.

Program 29: Find Last Occurrence of Character Write a program that finds the last position where character X appears.

Program 30: Find All Positions of Character Write a program that displays all positions where character X appears.

Program 31: Remove All Occurrences of Character Write a program that removes all occurrences of a specific character from the string.

Program 32: Remove Digits from String Write a program that removes all digits from the string.

Program 33: Remove Special Characters Write a program that removes all special characters, keeping only alphabets and spaces.

LEVEL 3: String Analysis & Comparison

Program 34: Check if Palindrome Write a program that checks if the string is a palindrome (reads same forwards and backwards). Example: "madam" → Yes.

Program 35: Check if Two Strings are Equal Write a program that compares two strings and checks if they are exactly equal (case-sensitive).

Program 36: Check if Two Strings are Equal (Ignore Case) Write a program that compares two strings ignoring case differences.

Program 37: Compare Two Strings Lexicographically Write a program that compares two strings and displays which comes first alphabetically.

Program 38: Check if String Contains Only Alphabets Write a program that checks if string contains only alphabetic characters.

Program 39: Check if String Contains Only Digits Write a program that checks if string contains only numeric digits.

Program 40: Check if String is Alphanumeric Write a program that checks if string contains only alphabets and digits (no special characters).

Program 41: Find Frequency of Each Character Write a program that displays frequency of each unique character in the string.

Program 42: Find Most Frequent Character Write a program that finds which character appears most frequently.

Program 43: Find Least Frequent Character Write a program that finds which character appears least frequently.

Program 44: Check if All Characters are Unique Write a program that checks if all characters in the string are unique (no repetition).

Program 45: Find First Non-Repeating Character Write a program that finds the first character that appears only once in the string.

Program 46: Find First Repeating Character Write a program that finds the first character that appears more than once.

Program 47: Count Distinct Characters Write a program that counts how many unique characters are in the string.

Program 48: Check if String Contains Substring Write a program that checks if a given substring exists in the main string.

LEVEL 4: String Manipulation

Program 49: Reverse String (In-Place) Write a program that reverses the string by modifying it.

Program 50: Reverse Each Word in String Write a program that reverses each word individually. Example: "Hello World" → "olleH dlroW".

Program 51: Reverse Word Order Write a program that reverses the order of words. Example: "Hello World" → "World Hello".

Program 52: Copy String to Another Write a program that copies one string to another without using strcpy().

Program 53: Concatenate Two Strings Write a program that concatenates (joins) two strings without using strcat().

Program 54: Extract Substring (Given Positions) Write a program that extracts substring from position start to end.

Program 55: Insert String at Position Write a program that inserts one string into another at a given position.

Program 56: Delete Characters from Position Write a program that deletes N characters starting from a given position.

Program 57: Replace Substring with Another Write a program that replaces all occurrences of a substring with another substring.

Program 58: Remove First Word Write a program that removes the first word from the string.

Program 59: Remove Last Word Write a program that removes the last word from the string.

Program 60: Extract First Word Write a program that extracts and displays only the first word.

Program 61: Extract Last Word Write a program that extracts and displays only the last word.

Program 62: Extract Nth Word Write a program that extracts the Nth word from the string (N given by user).

Program 63: Split String by Space Write a program that splits string into individual words and displays each word on new line.

Program 64: Join Array of Words into String Write a program that joins multiple words into a single string with spaces.

Program 65: Shift Characters Left by K Write a program that rotates string left by K positions. Example: "HELLO", K=2 → "LLOHE".

Program 66: Shift Characters Right by K Write a program that rotates string right by K positions.

Program 67: Swap First and Last Character Write a program that swaps the first and last characters of the string.

Program 68: Sort Characters in String (Ascending) Write a program that sorts all characters of string in alphabetical order.